Summary in Graph

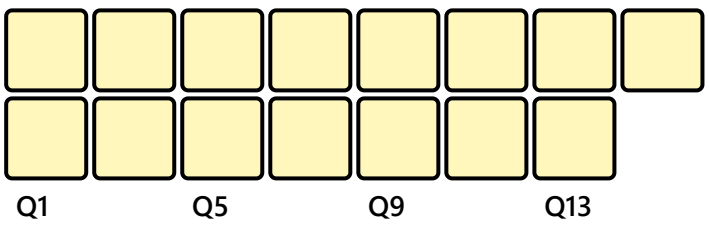
# Exam Summary (GO Classes GATE CS | Digital Logic | K Map | Weekly Quiz 2)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 15 (4 + 11) |
| Total Marks: | 26 (4 + 22) |
| Exam Duration: | 47 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (26) out of 264 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

# Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | X | | 1 |
| 01 | | 1 | | |
| 11 | | X | | |
| 10 | X | 1 | | X |

What will be the equation of the given $K$-map?

A. $A'B'D' + C'D + AB'C'$
B. $B'CD' + AB'C' + A'C'$
C. $B'D' + C'D$
D. $C'D + B'CD'$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The most simplified form of the Boolean function

$x(A, B, C, D) = \sum(7, 8, 9, 10, 11, 12, 13, 14, 15)$

(expressed in sum of minterms) is?

A. $A + A'BCD$

B. $AB + CD$

C. $A + BCD$

D. $ABC + D$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The boolean expression $\mathrm{AB} + \mathrm{AB}' + \mathrm{A}'\mathrm{C} + \mathrm{AC}$ is independent of the boolean variable

A. $\mathrm{A}$
B. $\mathrm{B}$
C. $\mathrm{C}$
D. None of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The switching expression corresponding to $f(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}) = \Sigma(1, 4, 5, 9, 11, 12)$ is

A. $BC'D' + A'C'D + AB'D$

B. $ABC' + ACD + B'C'D$

C. $ACD' + A'BC' + AC'D'$

D. $A'BD + ACD' + BCD'$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #5** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | 1 | | 1 |
| 01 | | | | |
| 11 | | | | |
| 10 | X | 1 | 1 | X |

Which will be the equation of simplification of the given $K$-map?

A. $AB' + B'CD' + A'B'C'$
B. $AB' + A'B'D' + A'B'C'$
C. $B'D' + AB' + B'C'$
D. $B'D' + A'B'C' + AB'$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #6** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Consider the following boolean function of four variables $f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$, the function is

A. Independent of one variable
B. Independent of two variables
C. Independent of three variables
D. Dependent on all variables

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #7** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The $K - map$ below defines a function $\mathrm{G(A, B, C, D)}$. Mark the answer below that corresponds to a minimal **SOP** expression for the function $\mathrm{G(A, B, C, D)}$ ?

**G**

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | x | 0 | 0 | x |
| 01 | 0 | 1 | 1 | 0 |
| 11 | x | 1 | 1 | 0 |
| 10 | 1 | 0 | 0 | 1 |

A. $\mathrm{BD + B'CD'}$

B. $B'D' + A'B'C + BD$

C. $BD + A'B'CD' + AB'C$

D. $B'D' + BD$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

| C | B | A | F |
|---|---|---|---|
| 0 | 0 | 0 | d |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | d |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

The truth table for implementing a Boolean variable $\mathrm{F}$ is given where $d$ represents don't care states. The minimized expression for $\mathrm{F}$ is:

A. $BC' + AC' + AB$
B. $AB + C'$
C. $A'BC' + AB'C' + ABC$
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following is the Boolean function for majority voting, assuming $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are inputs and $\mathrm{Y}$ is output?

a. $\mathrm{Y} = \mathrm{AB} + \mathrm{AC} + \mathrm{BC}$

b. $\mathrm{Y} = \mathrm{A} + \mathrm{B} + \mathrm{C}$

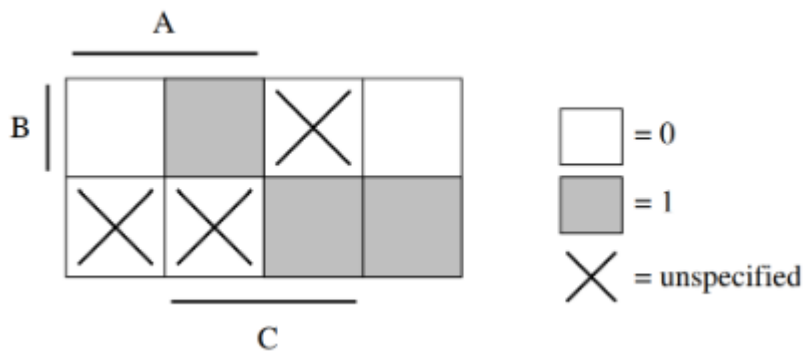
c. $\mathrm{Y} = \mathrm{ABC}$

d. $\mathrm{Y} = \mathrm{AB} + \mathrm{BC}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

A number of Boolean functions conform to the map shown below:




What is the minimum "sum of product" expression among the particular functions, having a prime implicant that can not appear in the minimum "sum of products" representation (that conform to the above map) ?

A. $\mathrm{B}' + \mathrm{C}$
B. $A'B' + AC$
C. $\mathrm{A}'\mathrm{B}' + \mathrm{B}'\mathrm{C}' + \mathrm{BC}$
D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Simplify Boolean function represented in sum of product of min-terms, $\mathrm{F}(x, y, z) = \Sigma(0, 2, 4, 5, 6)$

A. $z' + xy'$

B. $x'y'z' + xyz + xy'z'$

C. $xyz + x' + y' + z'$

D. $xy + yz + zx$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #12** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The simplified SOP (Sum of Product) from the Boolean expression

$$(\mathrm{P} + \overline{\mathrm{Q}} + \overline{\mathrm{R}}).(\mathrm{P} + \mathrm{Q} + \mathrm{R}) \,.\, (\mathrm{P} + \mathrm{Q} + \overline{\mathrm{R}})$$

is

A. $(\overline{\mathrm{P}}.\mathrm{Q} + \overline{\mathrm{R}})$
B. $(\mathrm{P} + \mathrm{Q}.\overline{\mathrm{R}})$
C. $(\mathrm{P}.\overline{\mathrm{Q}} + \mathrm{R})$
D. $(\mathrm{P}.\mathrm{Q} + \mathrm{R})$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #13** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The most simplified form of the Boolean function

$x(A, B, C, D) = \sum(7, 8, 9, 10, 11, 12, 13, 14, 15)$

(expressed in sum of minterms) is?

A. $A + A'BCD$

B. $AB + CD$

C. $A + BCD$

D. $ABC + D$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #14** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

\

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 1 | 0 |
| 01 | 0 | 1 | 0 | 0 |
| 11 | 1 | 0 | 1 | 0 |
| 10 | 0 | 1 | 0 | 1 |

The function shown in the figure when simplified will yield a result with _____ terms

A. $2$
B. $4$
C. $7$
D. $14$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #15** Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Find minimal POS (product of sum) expression for $F(a, b, c, d)$ given by the following K-map. Identify the Prime Implicants & Essential Prime Implicants clearly.

| cd \ ab | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | X | X | X |
| 01 | 1 | 1 | X | |
| 11 | 1 | | | |
| 10 | X | 1 | 1 | X |

Which of the following is/are correct statements about F?

A. Minimal POS: $(a' + b)\,(b' + c' + d')$
B. The number of Prime Implicants is $6$.
C. The number of Prime Implicants is $4$.
D. The number of Essential Prime Implicants is $1$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

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